MANGALORE INSTITUTE OF TECHNOLOGY AND ENGINEERING
(A Unit of Rajalaxmi Education Trust, Mangalore)
Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi Accredited by NAAC
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PROJECT PHASE –1 REPORT

"NEURO-ADAPTIVE CLOSED-LOOP SYSTEM FOR STROKE REHABILITATION USING REAL-TIME EEG AND COMPUTER VISION"

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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2025-26

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CERTIFICATE

This is to certify that the project work entitled **"NEURO-ADAPTIVE CLOSED-LOOP SYSTEM FOR STROKE REHABILITATION USING REAL-TIME EEG AND COMPUTER VISION"** is a bonafide work carried out by **CHAITHANYA S (4MT23AI015), CHAITHRANJALI ATHAVAR (4MT23AI016), NANDINI V C (4MT23AI035)** in partial fulfillment for the award of degree of **Bachelor of Engineering in ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** of **Mangalore Institute of Technology & Engineering (Autonomous)** during the year **2025-26**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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Rehabilitation using Real Time EEG and Computer Vision” would not have been possible without the support of those, whose constant guidance and encouragement helped turn our efforts into success.

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ABSTRACT

Stroke is a leading cause of long-term motor disability, often impairing voluntary movement due to damage in motor cortex regions. Conventional rehabilitation methods primarily focus on observable physical movement and lack mechanisms to assess underlying neural engagement, which is crucial for effective neuroplastic recovery. This limitation reduces the personalization and efficiency of therapy, especially in early stages where mental

effort may not translate into visible motion.

This project proposes a Neuro-Adaptive Closed-Loop Rehabilitation System that integrates real-time electroencephalography (EEG) and computer vision to bridge the gap between neural intent and physical execution. The system captures brain activity using EEG signals and extracts meaningful features such as Event-Related Desynchronization (ERD) to detect motor intent. Simultaneously, a vision-based module tracks upper-limb movements using pose estimation techniques to analyse joint trajectories and motion quality.

A key innovation of the system is the introduction of a Neuro-Motor Congruence Score (NMCS), which quantitatively aligns neural activation with physical movement. This enables the system to evaluate both mental effort and execution accuracy, overcoming the limitations of traditional performance-based scoring systems. Based on this score, a closed-loop feedback mechanism dynamically adjusts exercise difficulty, ensuring personalized and adaptive rehabilitation.

By combining neural decoding with motion analysis in a unified framework, the proposed system enhances therapy effectiveness, promotes neuroplasticity, and provides a low-cost, accessible solution for home-based stroke rehabilitation. This approach represents a significant step toward intelligent, patient-specific neurorehabilitation systems.

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INTRODUCTION

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1.INTRODUCTION

1.1 INTRODUCTION

Stroke is one of the leading causes of long-term disability worldwide, often resulting in partial or complete loss of motor function. Damage to motor cortex regions disrupts voluntary movement control, significantly affecting a patient's quality of life. Effective rehabilitation is critical for promoting motor recovery and restoring functional independence.

Traditional rehabilitation approaches rely on repetitive physical exercises guided by therapists or digital platforms. While these methods improve muscle strength and coordination, they primarily evaluate progress based on observable movement rather than neural activation.
However, motor recovery fundamentally depends on neuroplasticity — the brain's ability to reorganize and form new neural connections. Without sufficient cortical engagement, rehabilitation outcomes may remain limited.

Recent advancements in Brain-Computer Interface (BCI), electroencephalography (EEG), and machine learning have enabled real-time monitoring and interpretation of brain activity. These technologies open new possibilities for designing intelligent, closed-loop rehabilitation systems that adapt therapy based on neural engagement rather than just physical performance.

This project proposes a Neuro-Adaptive Rehabilitation System for stroke patients that utilizes real-time EEG analysis and machine learning techniques to dynamically adjust exercise difficulty based on motor cortex activation. The aim is to enhance personalized therapy and promote effective neuroplastic recovery through brain-driven adaptive intervention.

1.2 PROBLEM STATEMENT

Stroke causes motor impairment due to damage in motor cortex regions. Existing rehabilitation systems use generalized exercise protocols and evaluate progress based only on physical movement, without measuring neural engagement. Since motor recovery depends on active cortical participation and neuroplasticity, the absence of real-time brain monitoring limits therapy personalization and effectiveness. Therefore, there is a need for a closed-loop rehabilitation system that utilizes real-time EEG analysis and machine learning to adapt to exercise difficulty based on neural activation.

1.3 OBJECTIVES

The proposed project aims to develop a functional, real-time neuro-adaptive rehabilitation prototype by achieving the following milestones:

1. Establish a Multi-Modal Data Acquisition Pipeline: To synchronize live EEG streams (via LSL protocol) with real-time body tracking data using a standard webcam and MediaPipe Pose Estimation.
2. Implement Real-Time Neural Feature Extraction: To isolate Event-Related Desynchronization (ERD) in the Mu (8Hz - 13Hz) and Beta (13Hz-30Hz) bands as reliable indicators of a patient's "Mental Try.
3. Develop a Subject-Specific Intent Classifier: To design a lightweight Machine Learning model (such as a Linear Discriminant Analysis - LDA) that can be calibrated in 60 seconds to recognize the specific neural patterns of the user.
4. Construct a Neuro-Motor Congruence Scoring Engine: To create a mathematical "Match Score" that verifies if the patient’s physical arm movement (tracked by the camera) is happening at the exact same time as their neural intent (tracked by the EEG).
5. Design a Performance Accommodation Mechanism (PAM): To develop a logic-based loop that automatically makes the virtual exercise "easier" (requires less movement) if the system detects the patient is trying hard mentally but struggling physically.

1.4 MOTIVATION

Several converging factors make Adaptive Neuro-Rehab particularly timely and relevant.

- Stroke cases are increasing globally, creating a growing demand for effective and scalable rehabilitation solutions.
- Traditional rehabilitation methods lack personalization and do not utilize neural feedback for recovery.

- Advances in EEG technology, machine learning, and computer vision now enable real-time analysis of brain and movement data.
- There is a need for low-cost, home-based rehabilitation systems that reduce dependency on clinical setups.
- Integrating neural intent with physical movement can significantly improve neuroplasticity and recovery outcomes.
- Hence, this project aims to leverage modern technologies to build an adaptive, intelligent rehabilitation system.

1.5 FEASIBILITY STUDY

Technical Feasibility

The proposed system is technically feasible due to the availability of mature tools and frameworks for EEG signal processing, machine learning, and computer vision. Consumer-grade EEG devices enable real-time acquisition of neural signals, while established libraries such as MNE facilitate preprocessing and feature extraction (e.g., ERD). Computer vision frameworks like MediaPipe and OpenCV allow accurate real-time pose estimation using standard webcams. Machine learning models such as Linear Discriminant Analysis (LDA) can be efficiently implemented for intent classification with low computational overhead. The integration of these components into a unified pipeline is achievable using Python-based ecosystems, making real-time closed-loop processing practical on standard computing systems.

Operational Feasibility

The system is designed to be user-friendly and suitable for both clinical and home-based rehabilitation settings. Patients can use the system with minimal setup, involving an EEG headset and a webcam. The interface provides real-time feedback, enabling users to understand their performance without requiring constant supervision by a therapist. Additionally, the adaptive nature of the system allows it to adjust to individual patient capabilities, making it practical for diverse user groups, including those with severe motor impairments. This ensures ease of adoption and effective utilization in real-world scenarios.

Economic Feasibility

The proposed solution is cost-effective compared to traditional rehabilitation systems that rely on expensive robotic devices or clinical-grade equipment. By utilizing low-cost EEG headsets, standard webcams, and open-source software frameworks, the overall implementation cost is significantly reduced. The system eliminates the need for continuous clinical supervision, thereby lowering long-term rehabilitation expenses. This makes the solution economically viable and accessible, particularly for home-based therapy and resource-constrained environments.

Chapter - 2
LITERATURE SURVEY

Chapter - 2

2. LITERATURE SURVEY

2.1 LITERATURE REVIEW

To conduct the literature survey systematically, relevant research papers and journal articles related to EEG-based motor rehabilitation, BCI systems, signal processing, and machine learning techniques were reviewed. Reputable databases such as IEEE Xplore, Springer, ScienceDirect, and PubMed were used to gather recent and foundational studies. The reviewed works were analyzed to understand existing methodologies, limitations, and research gaps. This structured review helped establish a strong foundation for the proposed EEG-driven adaptive rehabilitation system.

S.No	Paper Title	Methodology	Outcome	Identification of Scope of Improvement	Year
1	Recommendations for Combining Brain-Computer Interface, Motor Imagery, and Virtual Reality in Upper Limb Stroke Rehabilitation: Qualitative Participatory Design Study.	Modeled Segmented Difficulty Scaling (SDS) logic for progressive cognitive loading. Utilized Qualitative Thematic Analysis to define task-logic variables.	Established 6 clinical design pillars, specifically advocating for Activity of Daily Living (ADL) based task modeling.	Does not provide a real-time automation algorithm for adjusting difficulty based on dual-stream (EEG+CV) data.	2025
2	A real time action scoring system for movement analysis and feedback in physical therapy using human pose estimation	Utilized MediaPipe Pose for 3D skeleton tracking. Calculated accuracy using Dynamic Time Warping (DTW) to measure the Euclidean distance between trajectories.	Demonstrated a high-precision real-time scoring engine (0–100) for physical therapy, enabling automated feedback.	Focuses purely on physical form; cannot detect Neural Intent, meaning it ignores the "mental effort" of paralyzed patients.	2025

3	A hybrid EMG–EEG interface for robust intention detection and fatigue-adaptive control of an elbow rehabilitation robot	Implemented Wavelet Packet Transform (WPT) for feature extraction and a Fuzzy Logic Controller to merge EEG/EMG signals for adaptive robot control.	Significant increase in intention detection accuracy; proved that Sensor Fusion reduces error rates caused by muscle fatigue.	EMG sensors are skin-contact dependent and bulky. Scope exists to replace EMG with non-contact Pose Estimation (MediaPipe).	2025
4	Adaptive LDA Classifier Enhances Real-Time Control of an EEG Brain–Computer Interface for Decoding Imagined Syllables	Developed an Adaptive-LDA (A-LDA) that updates the weight vector w and bias b based on the Stochastic Gradient Descent (SGD) of incoming trial data.	Proved that Online Adaptation maintains higher classification accuracy (>85%) over long sessions despite neural signal drift.	The model is tested for syllable decoding; requires adaptation for Complex Motor Movements in stroke rehabilitation.	2024

5	Towards Transforming Neurorehabilitation: The Impact of Artificial Intelligence on Diagnosis and Treatment of Neurological Disorders	Analyzed CNN and RNN (LSTM) architectures for time-series EEG. Evaluated Random Forest and SVM for biomarker-based recovery prediction.	Validated that AI-driven models can identify patient-specific recovery patterns, moving toward Personalized Precision Medicine.	Most models discussed are Computational-Heavy (Offline). Need for lightweight models for real-time home-rehab deployment.	2024
6	Characterization of Event Related Desynchronization in Chronic Stroke Using Motor Imagery Based Brain Computer Interface	Computed Event-Related Desynchronization (ERD) via: ERD%=RA−R ×100. Used Welch's Periodogram for spectral density estimation.	Mapped spatial shifts in EEG activation, confirming Ipsilateral activation as a compensatory mechanism in chronic stroke.	Fixed spatial filters (standard electrode caps) often fail to account for individual Cortical Reorganization.	2024

	for Upper Limb Rehabilitation				
7	Motor imagery brain–computer interface rehabilitation system enhances upper limb performance and improves brain activity in stroke patients: A clinical study	Applied Phase Lag Index (PLI) for functional connectivity analysis and correlated Mu-rhythm suppression with Fugl-Meyer (FM) clinical scores.	Proven that increased Neural Synchrony in the motor cortex is a direct bio-marker for significant motor function recovery.	Correlation is measured after the therapy; lacks a Real-Time Synchrony Metric to drive biofeedback during the task.	2023
8	Gamified Neurorehab ilitation Strategies for Post-stroke Motor Recovery: Challenges and Advantages	Integrated Behavioral Psychology Models with Reward-based Reinforcement Schedules within a game-engine framework.	Found that Gamification significantly increases patient compliance and "Time-on-Task," leading to better clinical outcomes.	Does not provide a Mathematical Difficulty Balancer; games often fail to adapt to the patient's immediate fatigue level.	2022
9	Toward an Adapted Neurofeedback for Post-stroke Motor Rehabilitation: State of the Art and Perspectives	Conducted a Systematic Review of Visual, Auditory, and Haptic feedback, analyzing Pooled Mean Differences via statistical forest plots.	Determined that Immersive Visual-VR feedback provides the highest "Sense of Agency" and triggers higher neural activation.	Most feedback is "Open-Loop"; it doesn't adjust based on the gap between the patient's intent and their actual physical ability.	2022
10	Effects of brain-computer interface-based training on post-stroke upper-limb rehabilitation: a meta-analysis	Evaluated Systematic Literature on Clinical BCI implementation, focusing on the transition from lab-based to Home-Based Systems.	Identified Hybrid Systems (combining sensors) and Cloud-based Processing as the future of scalable stroke rehab.	Highlights a lack of User-Friendly Platforms that combine clinical-grade accuracy with simple webcam/consumer-EEG hardware.	2025

11	A systematic review of the effects of brain-computer interface (BCI) on lower limb motor function, balance, and daily living in stroke patients	Conducted a meta-analysis of multiple clinical studies evaluating BCI impact on motor recovery, balance, and ADL. Synthesized quantitative and qualitative outcomes across datasets.	Provided strong meta-analysis evidence that BCI significantly improves lower limb motor function, balance, and daily living activities in stroke patients.	Lacks real-time adaptive frameworks; does not integrate multimodal signals (e.g., EEG + CV) for dynamic therapy personalization.	2026
12	Therapeutic effectiveness of brain-computer interfaces in stroke patients: systematic review	Reviewed clinical trials and experimental studies focusing on therapeutic effectiveness of BCI systems in rehabilitation.	Confirmed that BCI-based interventions improve motor recovery and rehabilitation outcomes in stroke patients.	Limited focus on scalability and real-time feedback systems; lacks integration with intelligent automation models.	2023
13	Efficacy and safety of brain–computer interface for stroke rehabilitation using non-invasive EEG	Evaluated non-invasive EEG-based BCI systems with emphasis on visual feedback mechanisms and safety analysis.	Demonstrated that EEG-based BCI systems are safe and effective, particularly when combined with visual feedback for motor recovery.	Focuses mainly on visual feedback; lacks multimodal interaction (e.g., haptics, AR/VR) and adaptive difficulty adjustment.	2025
14	Efficacy of kinesthetic motor imagery based brain computer interface for stroke subacute motor recovery	Used motor imagery (MI)-based BCI combined with external stimulation techniques for subacute stroke rehabilitation.	Showed improved motor recovery when MI-BCI is combined with stimulation techniques.	Does not incorporate real-time intention decoding refinement; limited personalization based on patient cognitive load.	2025

15	EEG-based brain network analysis of chronic stroke patients during BCI–FES rehabilitation training	Applied EEG-based brain network analysis during BCI combined with Functional Electrical Stimulation (FES) training.	Identified neural plasticity changes and improved brain connectivity during rehabilitation.	Focuses on analysis rather than real-time intervention; lacks automated decision-making for therapy adjustments.	2022

Table 2.1.1: Literature Survey

2.2 EXISTING SYSTEMS AND THEIR LIMITATIONS

System Type	Focus Area	Existing Limitations
Conventional Physiotherapy	Physical motor rehabilitation through therapist-guided exercises	Uses generalized exercise protocols; depends on therapist observation; lacks objective neural measurement; no real-time personalization based on brain activity.
Robotic-Assisted Rehabilitation Systems	Assisted limb movement using robotic exoskeletons or motorized devices	Focuses on mechanical movement execution; does not assess cortical engagement; high cost; limited adaptability to individual neural recovery.
Virtual Reality (VR)-Based Rehabilitation	Interactive motor training using immersive virtual environments	Evaluates task completion and movement accuracy; lacks real-time neural feedback; adaptation is performance-based not brain-based
EMG-Based Rehabilitation Systems	Muscle activity monitoring during rehabilitation	Measures peripheral muscle activation rather than central neural activity; cannot determine motor cortex engagement; limited insight into neuroplastic recovery
Brain-Computer Interface (BCI) Motor Imagery Systems	Detection of motor intent using EEG signals	Primarily designed for assistive communication or prosthetic control; limited integration with

		adaptive rehabilitation mechanisms
Fixed Protocol Digital Rehab Applications	Pre-programmed digital exercise platforms	Follow predefined therapy levels; lack patient-specific neural adaptation; do not incorporate real-time EEG-driven feedback

Table 2.2.1: Existing Systems and their limitations

2.3 PRIOR ART SEARCH

The literature survey highlights prior research efforts in neuroplasticity and rehabilitation frameworks. Key developments include*:*

- **Commercial BCI Systems (e.g., Recoverix, IpsiHand):**
  - Current State: Successfully uses Motor Imagery (MI) to trigger muscle stimulation (FES) or robotic orthotics using the 8–30 Hz Mu-rhythm.
  - The Gap: These are "Static Threshold" systems. They lack real-time difficulty adjustment based on neural fatigue and do not verify physical movement via external cameras (Computer Vision).
- **Adaptive Biofeedback Patents (e.g., US9827411B2, EP3458148A1):**
  - Current State: Protects the logic of "Intent-Triggered-Action" and adjusts visual feedback based on the Signal-to-Noise Ratio (SNR) of brainwaves.
  - The Gap: Adaptation is limited to the EEG signal alone. There is no Multi-Modal Data Fusion to cross-reference mental intent with physical joint trajectories (Pose Estimation).
- **AI-Driven Pose Estimation (e.g., MediaPipe-based Frameworks):**
  - Current State: Uses standard webcams to track joint angles and calculate Range of Motion (ROM) with high accuracy.
  - The Gap: These systems are "Brain-Blind." For a paralyzed patient, the system records a "zero" score because it cannot detect the Internal Mental Effort when physical execution fails

2.4 GAPS IDENTIFIED THROUGH LITERATURE SURVEY

1. **Lack of Multi-Modal Data Synchronization (The "Signal Silo" Gap)**

   - The Gap: Current rehabilitation systems typically function in silos—either tracking physical movement via cameras or monitoring brain activity via EEG. There is no real-time mathematical alignment between Neural Intent and Physical Execution.

   - The Impact: A patient can "cheat" a camera-based exercise using compensatory movements (e.g., using the shoulder to lift a paralyzed arm), and the system will incorrectly reward the movement without the corresponding neural activation.

2. **Absence of Neural-Driven "Partial Credit" (The "Binary Feedback" Gap):**

   - The Gap: Most AI-based scoring systems are performance-oriented. If a hemiplegic patient attempts to move but physically fails, standard pose-estimation systems record a "zero" score.

   - The Impact: This leads to "learned helplessness." Existing systems fail to detect and reward the Internal Mental Effort (Event-Related Desynchronization) when external execution is absent, which is critical for early-stage neuroplasticity.

3. **Static Thresholding and Lack of Adaptive Scaling (The "Fatigue" Gap):**

   - The Gap: Standard BCI and CV systems use fixed calibration baselines. They do not account for the non-stationary nature of neural signals or the physical fatigue that occurs during a 30-minute session.

   - The Impact: As a patient tires, their "Mental Effort" signal weakens. A static system will stop recognizing their intent, leading to frustration and session abandonment rather than adapting the difficulty to the patient's current state.

4. **High Technical Complexity and Cost Barriers (The "Accessibility" Gap):**

   - The Gap: Clinical-grade neuro-rehabilitation requires expensive 64-channel "wet" EEG caps and robotic exoskeletons that are difficult to set up without professional help.

- The Impact: This restricts effective therapy to high-end clinics. There is a clear lack of a "Zero-Contact" home-based solution that combines standard webcams with consumer-grade dry EEG headbands.

5. **Non-Congruent Visual Feedback (The "Agency" Gap):**

- The Gap: In many gamified systems, the virtual avatar moves based on pre-set timers or simple triggers, rather than being strictly contingent on the patient's simultaneous neural and physical "try."

- The Impact: For the brain to "rewire" itself, the visual feedback must be temporally congruent with the intent. Existing systems often lack this tight synchronization, reducing the effectiveness of the therapy.

Chapter – 3
SYSTEM REQUIREMENT SPECIFICATIONS

Chapter – 3

SYSTEM REQUIREMENT SPECIFICATION

3.1 OVERALL DESCRIPTION

This description offers a broad overview of the system, its goals, intended users, and the conditions in which it functions. It outlines what the system is designed to accomplish and why it is necessary, serving as a foundation before detailing the functional and non-functional requirements.

3.1.1 SYSTEM PERSPECTIVE

The proposed **Neuro-Adaptive Closed-Loop Rehabilitation System** is designed as a **standalone, multi-modal intelligent system** that integrates neural signal processing and computer vision-based motion analysis within a unified framework.

From a system perspective, the architecture follows a **closed-loop feedback paradigm**, where continuous input from multiple sensing modalities is processed in real time to generate adaptive outputs. The system operates at the intersection of three major domains:

- **Brain-Computer Interface (BCI)** → for decoding neural intent using EEG signals
- **Computer Vision** → for tracking physical movement using pose estimation
- **Machine Learning** → for classification, scoring, and adaptive decision-making

3.1.2 SYSTEM FUNCTIONS

The system performs a sequence of functions organized into a real-time processing pipeline. These functions ensure continuous monitoring, analysis, and adaptation of rehabilitation exercises.

1. Data Acquisition Function

   - Captures EEG signals from the user in real time using an EEG headset.
   - Simultaneously captures video input through a webcam for motion tracking.
   - Ensures synchronization between neural and visual data streams.

2. Signal Preprocessing Function

   - Filters EEG signals to remove noise and artifacts (e.g., eye blinks, muscle noise).
   - Applies band-pass filtering to isolate Mu (8–13 Hz) and Beta (13–30 Hz) frequency

bands.

- Normalizes and prepares data for feature extraction.

3. Feature Extraction Function

- Extracts neural features such as **Event-Related Desynchronization (ERD)** to represent motor intent.
- Extracts spatial features from pose estimation such as joint coordinates, angles, and trajectories.

4. Intent Classification Function

- Classifies EEG signals into categories (e.g., “intent present” vs “no intent”).
- Uses lightweight machine learning models (e.g., LDA) for real-time inference.

5. Motion Evaluation Function

- Computes movement accuracy using trajectory comparison techniques like **Dynamic Time Warping (DTW)**.
- Evaluates range of motion, smoothness, and correctness of movement.

6. Multi-Modal Fusion Function

- Combines neural intent (ERD) and motion accuracy (DTW score).
- Computes the **Neuro-Motor Congruence Score (NMCS)** as a unified metric.

7. Adaptive Feedback Function

- Provides real-time visual feedback to the user based on NMCS.
- Adjusts exercise difficulty dynamically using the Performance Accommodation Mechanism (PAM).
    - If neural effort is high but movement is low → reduce difficulty
    - If both are high → increase difficulty

8. Monitoring and Reporting Function

- Tracks user progress over time.
- Stores session data including EEG trends, movement performance, and NMCS scores.
- Allows clinicians to review progress and adjust therapy plans.

3.1.3 PRODUCT FUNCTIONS

The product exposes a set of high-level functional capabilities that define how users and external components interact with the system.

- **Rehabilitation & Data Acquisition:** Enables patients to perform exercises while capturing real-time EEG signals and motion data.
- **Neural Intent Detection:** Processes EEG signals to detect motor intent using feature extraction and classification.
- **Motion Tracking & Evaluation:** Uses computer vision to track body movement and evaluate accuracy using trajectory analysis.
- **Neuro-Motor Congruence Computation:** Integrates neural and physical metrics to compute the NMCS for performance assessment.
- **Adaptive Feedback & Control:** Dynamically adjusts exercise difficulty and provides real-time feedback based on user performance.
- **Monitoring & Reporting:** Records session data and enables progress tracking for users and clinicians.

3.1.4 USER CLASSES AND CHARACTERISTICS

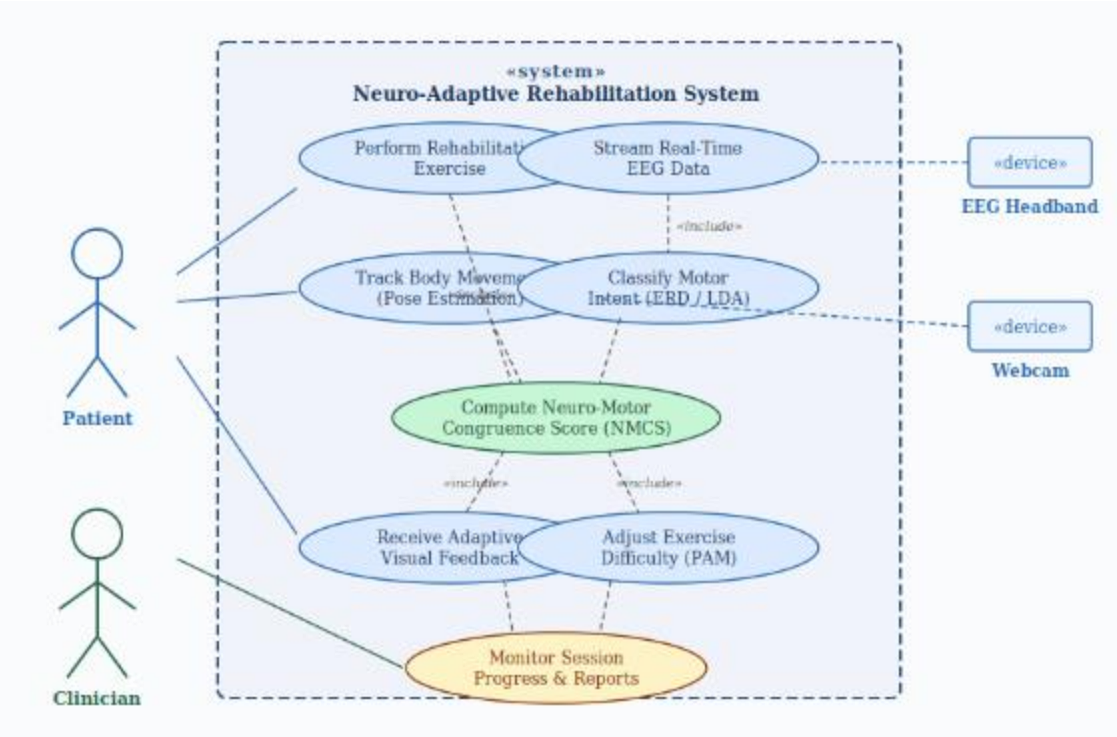

Figure 3.1.4: Use Case Diagram

The system serves three distinct user classes

1. **Patient (Primary User):** Individuals undergoing rehabilitation who interact directly with the system by performing exercises. They have limited technical expertise and require a simple, intuitive interface with real-time feedback.
2. **Clinician / Doctor (Secondary User):** Medical professionals who monitor patient performance, analyze reports (NMCS, progress trends), and guide therapy. They possess domain knowledge but may not have deep technical expertise.
3. **System Administrator / Developer:** Responsible for system setup, maintenance, calibration of EEG devices, and software updates. They have high technical expertise in AI, signal processing, and system integration.

3.1.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

- EEG Hardware Dependency: System performance is bounded by the signal quality of the consumer-grade EEG headband. Dry electrode systems typically exhibit higher impedance and susceptibility to motion artifacts compared to wet electrode clinical systems. Signal pre-processing (ICA artifact rejection, bandpass filtering) is essential to compensate.
- Real-Time Latency Constraint: The closed-loop feedback must operate within a maximum end-to-end latency of 500 ms (from neural event to feedback display) to maintain the temporal congruence required for neuroplastic learning. This imposes strict computational performance requirements on the classification and scoring modules.
- Dataset Scarcity: Publicly available, labeled EEG motor imagery datasets specific to chronic stroke populations are limited. PhysioNet Motor Imagery Dataset and BCI Competition IV Dataset 2a will be used for initial classifier training; subject-specific calibration will be essential.
- Inter-Subject Variability: EEG signals vary significantly between individuals due to differences in skull thickness, electrode placement, and neurological condition. The 60-second subject-specific LDA calibration is a design response to this constraint.
- Webcam Limitation: Standard 2D webcams cannot capture depth information, limiting the precision of 3D joint angle computation. MediaPipe's 3D landmark estimation partially compensates through model-based inference.

3.1.6 ASUMPTIONS AND DEPENDENCIES

- It is assumed that the patient's EEG headband (e.g., OpenBCI Cyton) is correctly placed with electrodes covering C3, C4, and Cz positions (sensorimotor cortex) per the International 10-20 system.
- It is assumed that the patient can perform at least Motor Imagery of upper limb movement (imagined movement), even if physical execution is impossible. Patients in a vegetative or minimally conscious state are not within the scope of this system.
- The system depends on the Lab Streaming Layer (LSL) protocol for EEG data streaming. The EEG headband must support LSL-compatible data output or use an intermediate driver such as OpenBCI GUI.
- The system depends on the availability of Python 3.9+, the MNE library (version 1.4+), MediaPipe (version 0.10+), and scikit-learn (version 1.2+) in the deployment environment.
- Adequate ambient lighting is assumed for effective webcam-based pose estimation. Sessions conducted in poorly lit environments may degrade CV accuracy.

3.2 SPECIFIC REQUIREMENTS

3.2.1 HARDWARE REQUIREMENTS

The proposed Neuro-Adaptive Rehabilitation System requires the following hardware components to support real-time EEG acquisition, motion tracking, and processing:

S. No.	Component	Specification
1	Processing Unit (CPU)	Intel i5 / AMD Ryzen 5 or higher
2	GPU (Optional)	NVIDIA GTX/RTX series (for faster ML and vision processing)
3	Memory (RAM)	Minimum 8 GB (16 GB recommended)
4	EEG Device	Consumer-grade EEG headset (dry electrodes, LSL support preferred)
5	Camera (Webcam)	HD webcam (minimum 720p, preferably 1080p)

6	Storage	Minimum 256 GB (SSD preferred for faster access)
7	Display Unit	Monitor for real-time feedback visualization
8	Input Devices	Keyboard and mouse for system interaction

Table 3.2.1: Hardware Requirements Specifications

3.2.2 SOFTWARE REQUIREMENTS

The proposed Neuro-Adaptive Rehabilitation System requires the following hardware components.

Software Component	Package / Version	Purpose
Programming Language	Python 3.9 – 3.11	Core development language for all modules
EEG Processing	MNE-Python 1.4+	Raw EEG reading, filtering, ICA, ERD computation
Signal Streaming	pylsl (Lab Streaming Layer) 1.16+	Real-time EEG data acquisition from headband
Computer Vision	MediaPipe 0.10+ / OpenCV 4.8+	Real-time pose estimation and joint tracking
Machine Learning	scikit-learn 1.2+ / TensorFlow 2.12+ (optional for EEGNet)	LDA classifier, cross-validation, EEGNet deep learning
Scientific Computing	NumPy 1.24+, SciPy 1.10+	Signal processing, DTW computation, statistical analysis
Data Visualization	Matplotlib 3.7+ / PyQtGraph	Real-time waveform display, session dashboards
GUI Framework	PyQt5 / Tkinter	Patient-facing graphical interface
Data Storage	SQLite 3.40+ / CSV	Session log storage, ERD and score history
Dataset (Training)	BCI Competition IV 2a, PhysioNet EEG Motor Movement/Imagery Dataset	Offline classifier pre-training before subject-specific calibration

Table 3.2.2: Software Requirement Specifications

3.2.3 FUNCTIONAL REQUIREMENTS

The functional requirements define the core operations of the system, describing how inputs are processed to generate outputs at each stage of the pipeline.

1. **EEG Data Acquisition**

   The system shall acquire raw EEG data from the connected headband via LSL protocol at a minimum sampling rate of 250 Hz, with timestamp synchronization accurate to within 5 ms.

2. **EEG Preprocessing**

   The system shall apply a 1–40 Hz bandpass filter, a 50 Hz notch filter (power line noise rejection), and Independent Component Analysis (ICA) for ocular artifact removal to raw EEG streams in real time.

3. **ERD Feature Extraction**

   The system shall compute Event-Related Desynchronization (ERD) in the Mu (8–13 Hz) and Beta (13–30 Hz) bands using Welch's Periodogram or STFT within a sliding 2-second window, updating every 250 ms.

4. **Motor Intent Classification:**

   The system shall classify each 250 ms EEG epoch as either 'Motor Imagery Active' or 'Rest' using a subject-specific LDA classifier with a target accuracy >80%.

5. **Pose Estimation**

   The system shall track 33 upper body landmarks using MediaPipe Pose Estimation at a minimum of 25 fps, computing elbow flexion angle, shoulder abduction angle, and wrist trajectory in real time.

6. **Movement Scoring**

   The system shall compute a DTW-based Accuracy Score (0–100) for each exercise repetition by comparing the patient's joint trajectory against a pre-recorded reference trajectory for the target exercise.

7. **NMCS Computation**

   The system shall compute the Neuro-Motor Congruence Score (NMCS) as a weighted composite of the ERD-based Neural Intent Score (NIS) and the DTW-based Physical Execution Score (PES), updated per repetition.

8. **Adaptive Difficulty (PAM)**

   The system shall automatically reduce the target Range of Motion (ROM) threshold for the next repetition by 15% if the NMCS indicates high neural intent (NIS > 0.7) combined with low physical execution (PES < 0.4).

9. **Real-Time Feedback Display**

   The system shall display the current NIS, PES, and NMCS values to the patient as a real-time dashboard with visual progress indicators, updating within 500 ms of each event.

10. **Session Report Generation**

    The system shall generate a PDF session report at the end of each rehabilitation session, including time-series plots of ERD, movement accuracy, NMCS trend, and a session summary table.

3.2.4 NON- FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality attributes of the system rather than specific features. They define how well the system should perform.

- **Performance:** The system shall process EEG and video data in real time with low latency to ensure smooth feedback.
- **Reliability:** The system shall operate consistently with minimal errors during continuous usage.
- **Usability:** The interface shall be simple and user-friendly for patients with limited technical knowledge.
- **Accuracy:** The system shall provide accurate detection of motor intent and movement evaluation.
- **Robustness:** The system shall handle noisy EEG signals and varying environmental conditions effectively.

Chapter – 4
GANTT CHART

Chapter – 4

GANTT CHART

4.1 GANTT CHART
2026

ACTIVITY/MONTH	**FEB**	**MAR**	**APR**	**MAY**	**JUN**	**JUL**	**AUG**	**SEP**	**OCT**	**NOV**
SYNOPSIS										
PRESENTATION IDEA										
SYSTEM REQUIREMNT SPECIFICATION										
SYSTEM ARCHITECTURE										
IMPLEMENTATION										
PRESENTATION ON WORK PROGRESS										
TESTING										
RESULT AND REPORT										

Chapter – 5
SYSTEM ARCHITCTURE AND METHODOLOGY

Chapter – 5

SYSTEM ARCHITECTURE AND METHODOLOGY

5.1 OVERVIEW

The proposed Neuro-Adaptive Closed-Loop Rehabilitation System is a multimodal, brain-driven therapeutic platform designed to address the fundamental limitations of existing stroke rehabilitation paradigms. Unlike conventional systems that evaluate recovery solely through observable physical movement, this system integrates two orthogonal sensing modalities — electroencephalography (EEG) and computer vision to achieve a complete, bidirectional picture of a patient's recovery state.

At its core, the system continuously acquires cortical neural signals from the patient's motor cortex through a consumer-grade, 14-channel dry EEG headset. These signals are streamed in real-time over the Lab Streaming Layer (LSL) protocol to a processing pipeline that isolates event-related desynchronization (ERD) in the Mu (8–13 Hz) and Beta (13–30 Hz) frequency bands the established electrophysiological correlates of motor intent and effort. Simultaneously, a standard RGB webcam captures the patient's arm movements, which are processed through the MediaPipe Pose Estimation framework to extract 3D joint landmarks and compute a Dynamic Time Warping (DTW)-based movement accuracy score.

The two data streams are fused inside a Neuro-Motor Congruence Scoring Engine (NMCSE), which produces a composite score reflecting both the patient's neural effort and physical performance. This composite score feeds into a Performance Accommodation Mechanism (PAM), a rules-based adaptive controller that dynamically adjusts the difficulty of on-screen rehabilitation exercises in real-time. When a patient attempts to move but fails physically, the PAM detects the residual cortical activation and still rewards the mental effort, preventing learned helplessness and actively promoting neuroplasticity.

The system is designed for low-cost deployment in home environments, requiring only a standard laptop, a consumer EEG headset, and a webcam, thereby democratizing access to neuro-adaptive rehabilitation outside expensive clinical settings.

5.2 ARCHITECTURE DESIGN

The system architecture follows a layered, closed-loop design organized into five functional tiers: (i) Sensing Layer, (ii) Signal Processing Layer, (iii) Intelligence Layer, (iv) Fusion & Scoring Layer, and (v) Feedback & Adaptation Layer.

5.2.1 HIGH LEVEL SYSTEM ARCHITECTURE

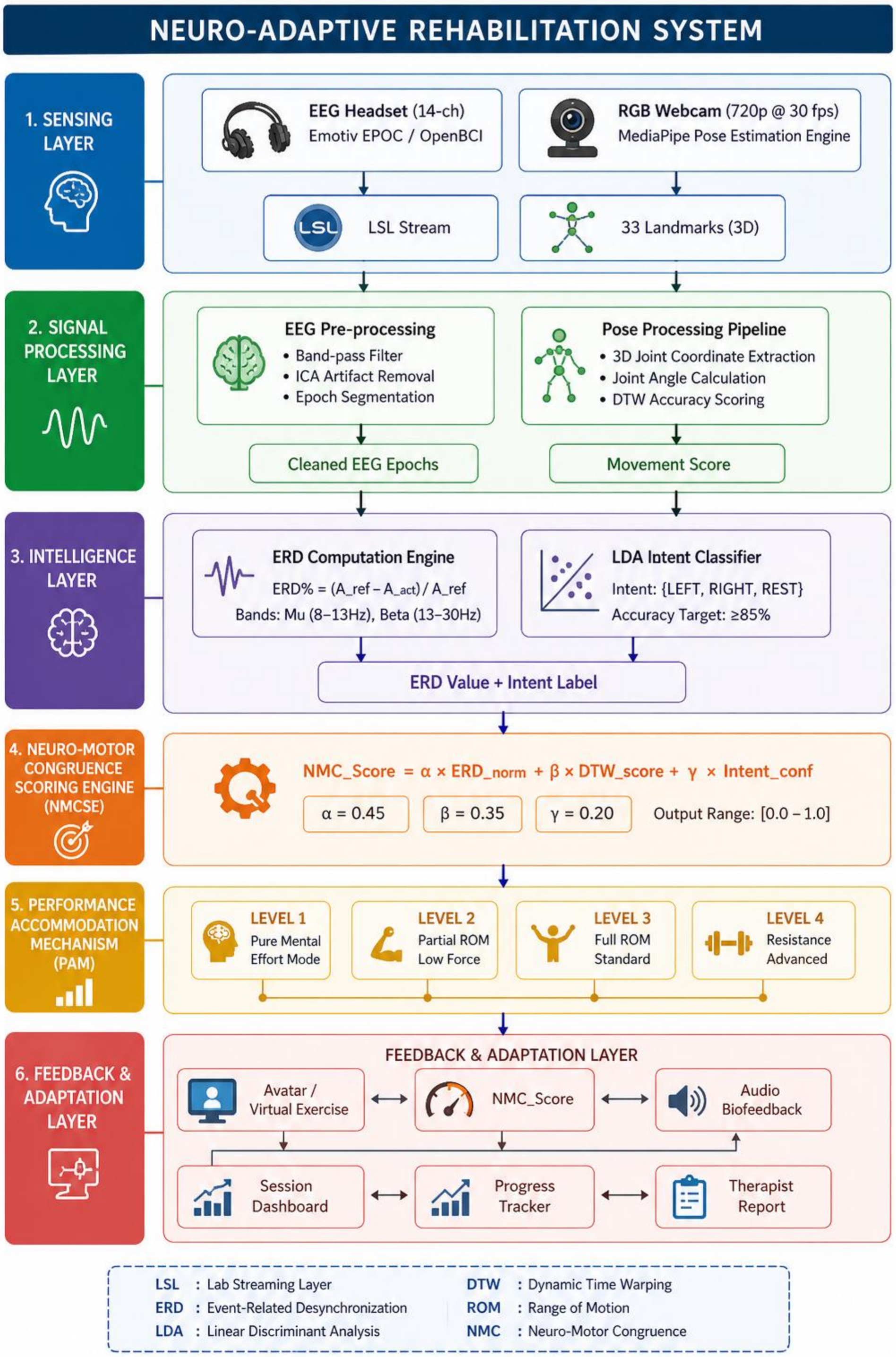

Figure 5.2: High-level block diagram of the NARS showing all Functional tiers

5.2.2 COMPONENT INTERACTION DESCRIPTION

Layer	Key Component	Input	Output
Sensing	Emotiv EPOC+ EEG / OpenBCI	Brain electrical potentials (µV)	Raw 14-ch EEG @ 256 Hz; LSL stream
Sensing	RGB Webcam + MediaPipe	Pixel frames @ 30 fps	33 × (x,y,z) joint coordinates
Signal Processing	EEG Pre-processing Module	Raw EEG epochs (2 s windows)	Cleaned ERD-featured epochs
Signal Processing	Pose DTW Scorer	3D landmark stream	DTW accuracy score ∈ [0,100]
Intelligence	ERD Computation Engine	Filtered EEG power spectra	ERD% in Mu and Beta bands
Intelligence	LDA Intent Classifier	ERD feature vector	Intent label + confidence ∈ [0,1]
Fusion	NMCSE	ERD norm, DTW score, Intent conf.	NMC_Score ∈ [0.0, 1.0]
Adaptation	PAM Controller	NMC_Score + session history	Difficulty level {1,2,3,4}
Feedback	Visual / Audio Feedback	Difficulty level + NMC_Score	Adaptive exercise UI; audio cue

Table 5.2.2: Layer-wise component interaction matrix for the proposed system architecture.

5.3 DATA DESIGN

5.3.1 DATA COLLECTION AND SOURCES

The system utilizes a dual-stream data acquisition architecture. Each stream is independently timestamped and later synchronized using a shared LSL master clock.

A. EEG Data Acquisition

EEG signals are recorded using a 14-channel consumer-grade dry EEG headset (e.g., Emotiv EPOC+) at a sampling rate of 256 Hz. The 14 active electrodes (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) are positioned according to the International 10-20 system. For the purposes of motor imagery classification, electrodes overlying the sensorimotor cortex (C3, Cz, C4 positions, approximated by FC5, FC6, and their neighbours) are prioritized.

Data streams are transmitted via the LSL (Lab Streaming Layer) protocol over a local network, providing microsecond-precision timestamps essential for multi-modal synchronization. The dataset used for model training and validation is the publicly available BCI Competition IV Dataset 2a (Graz dataset), which contains EEG recordings from 9 subjects performing four classes of motor imagery tasks (left hand, right hand, feet, tongue) at 250 Hz, 22-channel.

B. Computer Vision Data Acquisition

The computer vision stream is captured through a standard USB RGB webcam operating at 720p resolution at 30 fps. The Google MediaPipe Pose Estimation framework (v0.10) is employed to extract a 33-landmark, 3D skeletal pose model from each video frame. For upper-limb rehabilitation, the key landmarks of interest are: shoulder (L/R), elbow (L/R), wrist (L/R), and finger tips, providing joint angle vectors $\theta = [\theta\_shoulder, \theta\_elbow, \theta\_wrist]$.

Reference exercise trajectories are pre-recorded by a certified physiotherapist and stored as template time-series. These templates serve as the ground-truth movement patterns against which patient performance is scored using DTW.

C. Dataset Summary

Source	Modality	Subjects / Samples	Purpose

BCI Comp. IV Dataset 2a	EEG (22-ch, 250 Hz)	9 subjects, 288 trials each	LDA classifier training & validation
Physionet EEG-MMI	EEG (64-ch, 160 Hz)	109 subjects, 14 runs	Cross-dataset generalization test
Therapist-recorded Templates	RGB Video (30 fps)	5 reference exercises	DTW baseline trajectory library
Live Patient Session	EEG + Video (dual-stream)	Per-session acquisition	Real-time inference & adaptive control

5.3.2 DATA PRE-PROCESSING

Pre-processing is performed independently on each modality before feature extraction and fusion. The goal is to maximize the signal-to-noise ratio (SNR) while minimizing latency, given the real-time operating constraint.

A. EEG Pre-processing Pipeline

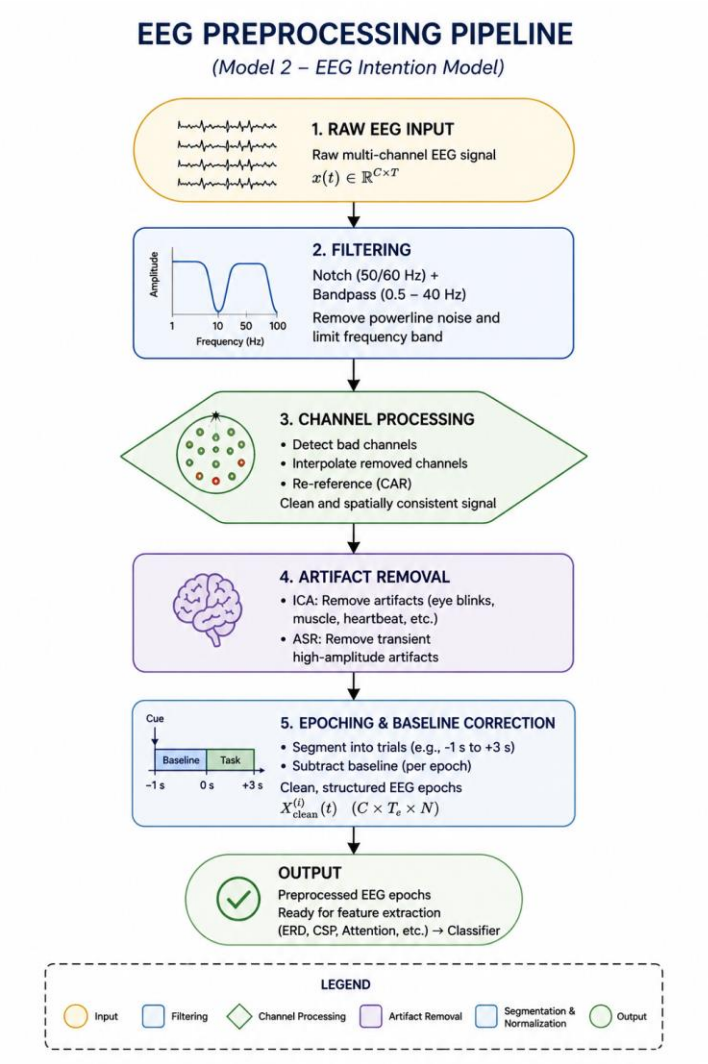

Figure 5.3.2 (i): EEG Pre-processing Pipeline

B. Pose Data Pre-processing

Landmark Filtering: A one-euro filter is applied to each (x,y,z) landmark coordinate to suppress high-frequency jitter inherent in single-frame MediaPipe predictions.

Coordinate Normalization: All landmark positions are normalized relative to the midpoint of the two hip landmarks to ensure scale and position invariance across different patient body sizes.

Joint Angle Extraction: Joint angles are computed from three-point landmark triplets using the law of cosines: $\theta = \arccos[(v_1 \cdot v_2) / (|v_1| \times |v_2|)]$, where v_1 and v_2 are bone vectors.

Temporal Resampling: The pose angle time-series is resampled to a canonical length of 100 frames per exercise repetition to enable fair DTW comparison across varying execution speeds.

5.3.3 FEATURE ENGINEERING/ SELECTION

A. EEG FEATURE ENGINEERING

The electroencephalography (EEG) feature extraction process is designed to capture spectral, spatial, temporal, asymmetry, and cognitive characteristics associated with motor intention and attention. These features are derived from preprocessed and segmented EEG signals $X^{(i)}(t) \in R^{C \times T}$ where C denotes the number of channels and T represents the number of temporal samples per epoch.

1. Spectral Features

Spectral features quantify the distribution of signal power across frequency bands relevant to motor activity. Specifically, band power features are computed in the mu $(8 - 12Hz)$ and beta $(13 - 30Hz)$ frequency bands. The band power is defined as:

$$P_f = \frac{1}{T} \sum_{t=1}^{T} \left| x_f(t) \right|^2$$

where $x_f(t)$ represents the bandpass-filtered EEG signal in frequency f .

Event-Related Desynchronization (ERD) is computed to quantify the relative decrease in band power with respect to a baseline period. It is defined as:

$$ERD(t) = \frac{P_b(t) - P(t)}{P_b(t)} \times 100$$

where $P_b(t)$ denotes the baseline power and $P(t)$represents the instantaneous band power.

2. Spatial Features

Spatial features aim to enhance discriminability between classes by exploiting spatial patterns across EEG channels. Common Spatial Patterns (CSP) are employed to obtain spatially filtered signals:

$$Z = W^T X$$

where Z_i represents the i -th CSP component.

3. Temporal Features

Temporal features capture the dynamic evolution of ERD over time within each epoch. The mean ERD is computed as:

$$\mu_{ERD} = \frac{1}{T}\sum_{t=1}^{T} ERD(t)$$

The peak ERD is defined as the maximum value of the ERD signal:

$$\max_t ERD(t)$$

The latency to peak ERD is given by:

$$t_{peak} = \arg\max_t ERD(t)$$

Additionally, the rate of change of ERD, or ERD slope, is computed as the temporal derivative:

$$\frac{d}{dt} ERD(t)$$

4. Asymmetry Features

Asymmetry features capture lateralization effects in motor cortex activation. A commonly used metric is the hemispheric difference between channels located over the left and right motor cortices:

$$A = ERD_{C3} - ERD_{C4}$$

where ERD_{C3} and ERD_{C4} correspond to ERD values at electrodes C3 and C4, respectively.

5. Attention Features

Cognitive engagement is quantified using attention-related spectral features. Frontal theta power is computed as:

$$P_\theta = \frac{1}{T}\sum_{t=1}^{T} |x_\theta(t)|^2$$

where $x_\theta(t)$ represents the EEG signal in the theta band.

Alpha suppression, which reflects attentional engagement, is defined as:

$$\Delta P_\alpha = P_{baseline} - P_\alpha(t)$$

where $P_\alpha(t)$ is the alpha band power during the task and $P_{baseline}$ is the baseline alpha power.

B. **Motion Feature Engineering**

Motion features are extracted from joint trajectories obtained via pose estimation and represent kinematic, angular, smoothness, symmetry, and temporal alignment characteristics of movement.

1.Kinematic Features

Kinematic features describe the position and motion of joints over time. The joint position at time t is defined as:

$$p_t = (x_t, y_t, z_t)$$

Velocity is computed as the first-order temporal derivative of position:

$$v_t = \frac{p_t - p_{t-1}}{\Delta t}$$

Acceleration is defined as the derivative of velocity:

$$a_t = \frac{v_t - v_{t-1}}{\Delta t}$$

2.Angular Features

Angular features capture joint configurations and movement range. The joint angle formed by three points a , b , and c is computed as:

$$\theta = \cos^{-1}\left(\frac{(a-b)\,.\,(c-b)}{\left||a-b|\right|\,\left||c-b|\right|}\right)$$

The range of motion (ROM) is defined as the difference between maximum and minimum joint angles over time:

$$ROM = \max(\theta_t) - \min(\theta_t)$$

3.Smoothness Feature:

Smoothness of motion is evaluated using higher-order derivatives. Jerk, defined as the rate of change of acceleration, is computed as:

$$j_t = a_t - a_{t-1}$$

Velocity variance is used as a measure of movement stability:

$$\sigma_v^2 = var(v_t)$$

4.Symmetric Features:

Symmetry features assess differences between bilateral movements. The Euclidean distance between corresponding joints on the left and right sides is defined as:

$$S = \left\| p_{left} - p_{right} \right\|$$

5.DTW - Based Features:

Dynamic Time Warping (DTW) is used to evaluate similarity between patient movement sequences Q and reference sequences R . The DTW distance is defined as:

$$DTW(Q, R) = \min_{\pi} \sum_{(i,j)\in\pi} d(q_i, r_j)$$

where π represents the optimal alignment path.

The path length $|\pi|$ represents the complexity of the alignment.

5.4 MODEL/MODULE DESIGN

The proposed system is composed of three primary computational modules: (i) Motion Quality Assessment Model, (ii) EEG-Based Intention Detection Model, and (iii) Adaptive Recommendation Model. These modules operate on heterogeneous data streams, namely visual pose sequences and multichannel EEG signals, and are integrated through a Neuro-Motor Congruence Score (NMCS) framework. Each module is mathematically formulated to ensure interpretability while maintaining robustness to temporal variability and signal noise.

5.4.1 MOTION QUALITY ASSESEMENT MODEL

The motion quality assessment model evaluates the correctness and consistency of performed rehabilitation exercises by comparing patient-generated joint trajectories with predefined reference trajectories. The model accounts for both spatial configuration and temporal dynamics of movement.

1. Pose Representation and Trajectory Construction:

Let the sequence of joint positions extracted from pose estimation be defined as:

$$Q = \{q_1, q_2, \dots, q_n\}\ ,\ q_i \in\ R^d$$

where d denotes the dimensionality of the joint representation (e.g., 2D or 3D coordinates). Similarly, the reference trajectory is denoted as:

$$R = \{r_1, r_2, \dots, r_m\}$$

2. Local Distance Metric

The similarity between two frames q_i and r_i is computed using a weighted distance function incorporating both position and velocity:

$$d(q_i, r_j) = w_p \parallel p_i - p_j \parallel^2 + w_v \parallel v_i - v_j \parallel^2$$

where p_i and v_i denote position and velocity vectors, respectively, and w_p, w_v are weighting coefficients.

3. Dynamic Time Wrapping (DTW):

Temporal alignment between Q and R is performed using Dynamic Time Warping:

$$DTW(Q,R) = \min_{\pi} \sum_{(i,j)\in\pi} d\left(q_i, r_j\right)$$

where π represents the optimal warping path subject to constraints such as the Sakoe–Chiba band.

4. Motion Score:

The DTW distance is transformed into a normalized motion quality score:

$$S_{phys} = \exp\left(-\lambda \cdot DTW(Q,R)\right)$$

Where, λ - Scaling parameter.

5.4.2 EEG INENTION MODEL:

The EEG intention model detects motor intent and cognitive engagement from preprocessed EEG signals.

1. Signal Representation:

$$X^{(i)}(t) \in R^{C\times T}$$

where C is the number of channels and T is the number of temporal samples.

2. Feature Extraction:

- Spectral Features:

  Band power is computed as:

$$P_f = \frac{1}{T}\sum_{t=1}^{T}\left|x_f(t)\right|^2$$

Event-Related Desynchronization (ERD) is defined as:

$$ERD(t) = \frac{P_b(t) - P(t)}{P_b(t)} \times 100$$

- Spatial Features (CSP):

Spatial filtering is applied using Common Spatial Patterns:

$$Z = W^T X$$

Feature vectors are derived from log-variance:

$$f_i = \log(var(Z_i))$$

- Temporal Features:

  Temporal characteristics of ERD are captured as:

$$\mu_{ERD} = \frac{1}{T} \sum_{t=1}^{T} ERD(t)$$

$$t_{peak} = \arg\max_t E\,RD(t)$$

3. Classification:

The extracted feature vector z_t is input to a classifier:

$$\hat{y_t} = f_{SVM}(z_t)$$

where f_{SVM} denotes a support vector machine. The classifier outputs probabilities corresponding to movement intention and attention states.

The neural engagement score is then defined as:

$$S_{neural} = \sigma\left(\frac{ERD - \mu}{\sigma}\right)$$

where $\sigma(.)$is the sigmoid function.

5.4.3 NEURO MOTOR CONGRUENCE FUSION

The outputs of the motion and EEG models are fused to compute the Neuro-Motor Congruence Score (NMCS)

1. Adaptive Weighting:

$$\alpha = \frac{1}{1 + e^{-\beta(\tau - S_{phys})}}$$

where β controls sensitivity and τ is a performance threshold.

2. Temporal Consistency:

$$C = \exp\left(-\frac{(t_{intent} - t_{movement})^2}{2\sigma^2}\right)$$

3. NMCS Computation:

$$NMCS = C\left[\alpha S_{neural} + (1-\alpha)S_{phys}\right]$$

5.4.4 ADAPTIVE RECOMMENDATION MODEL:

The adaptive recommendation model dynamically adjusts exercise difficulty based on NMCS.

1. State Update:

$$v_{t+1} = \gamma v_t + \eta(NMCS_t - \tau)$$

$$D_{t+1} = D_t + v_{t+1}$$

where γ is a momentum term and η is the learning rate.

2. Decision Policy:

The system generates recommendations based on the updated difficulty level:

$$R_t = \{Increasedifficulty D_{t+1} > u\ , Maintain l \leq D_{t+1} \leq u\ , Decreasedifficulty D_{t+1} < l\}$$

5.5 DATA FLOW DESIGN

The data flow architecture of the proposed system is modeled using a hierarchical Data Flow Diagram (DFD) representation to clearly illustrate the transformation of inputs into system outputs across multiple levels of abstraction. The design follows a three-level decomposition approach, consisting of Level 0 (context diagram), Level 1 (system decomposition), and Level 2 (process-level breakdown). Each level progressively refines the system structure and data interactions.

5.5.1 DFD LEVEL 0: CONTEXT DIAGRAM

At the highest level of abstraction, the system is represented as a single processing entity, namely the Neuro-Adaptive Rehabilitation System, which interacts with external entities including the patient, therapist, and clinical database.

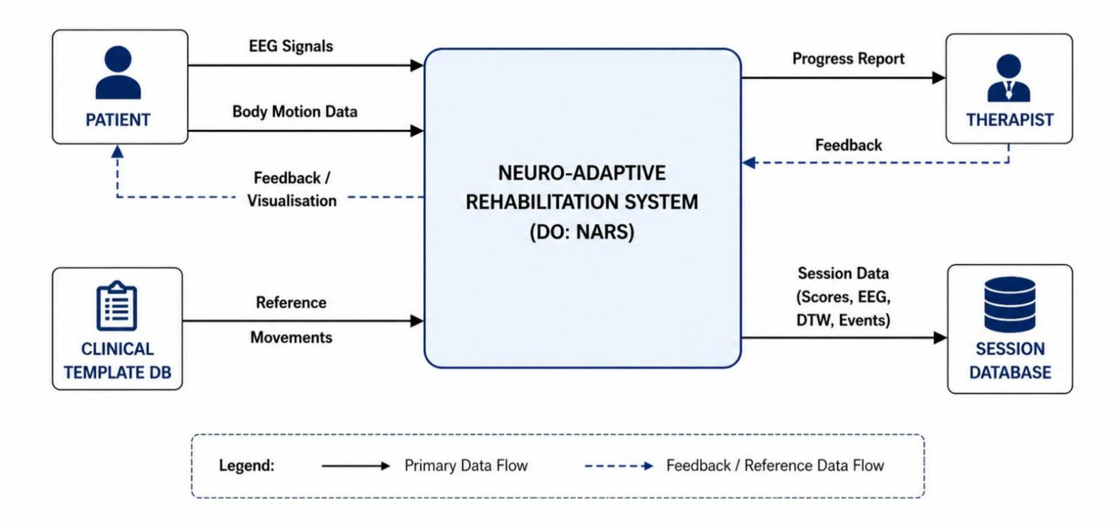

Figure 5.5.1: DFD Level 0 – Context Diagram

The patient provides two primary input streams: EEG signals and body motion data. These inputs are processed within the system to generate rehabilitation metrics and feedback. The system outputs include progress reports for the therapist and session data stored in a centralized database. Additionally, a clinical template database provides reference movement patterns used for motion comparison.

This level defines the **system boundary**, external entities, and primary data exchanges without exposing internal processing details.

5.5.2 DFD LEVEL 1: SYSTEM DECOMPOSITION

At Level 1, the system is decomposed into major functional processes that correspond to the core modules of the architecture.

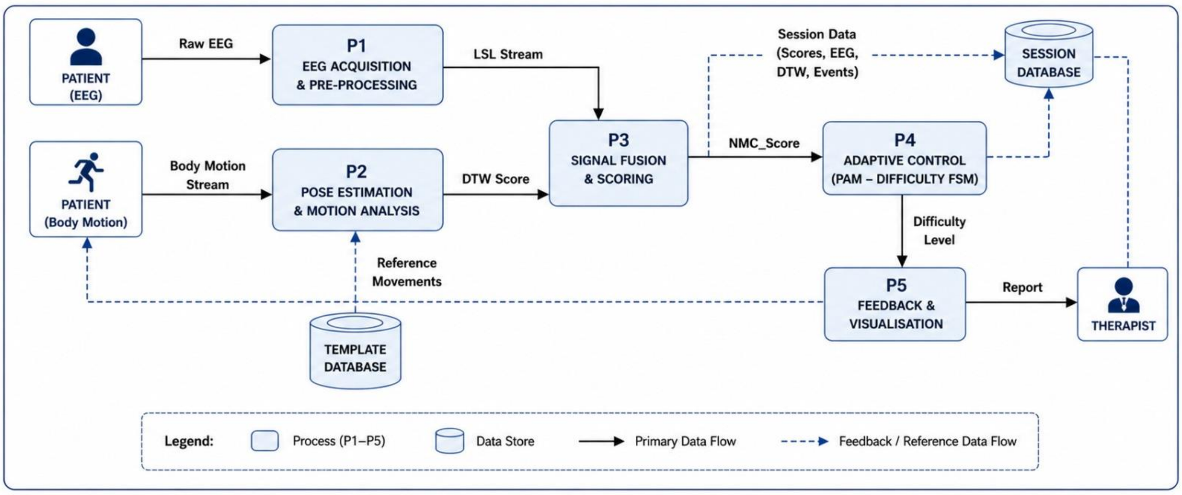

Figure 5.5.2: DFD Level 1 – System Decomposition

The system is divided into the following processes:

- **P1: EEG Acquisition and Pre-processing**
  This module receives raw EEG signals and performs filtering, artifact removal, and epoch segmentation to generate clean EEG data suitable for feature extraction.
- **P2: Pose Estimation and Motion Processing**
  Body motion input is processed to extract joint trajectories, which are compared against reference templates using DTW-based scoring.
- **P3: Signal Fusion and Scoring**
  Outputs from EEG processing and motion analysis are combined to compute the Neuro-Motor Congruence Score (NMC Score).
- **P4: Adaptive Control (PAM)**
  The computed NMC score is used to dynamically adjust exercise difficulty through a finite-state mechanism.
- **P5: Feedback and Visualization**
  The system generates real-time feedback for the patient and stores session data for therapist review.

Data stores include the **Template Database** (reference movements) and **Session Database** (scores, EEG data, and events). The data flow between modules ensures a continuous loop from sensing to feedback.

5.5.3 DFD LEVEL- 2: EEG PREPROCESSING DECOMPOSITION(P1)

Level 2 provides a detailed breakdown of the EEG processing pipeline within Process P1.

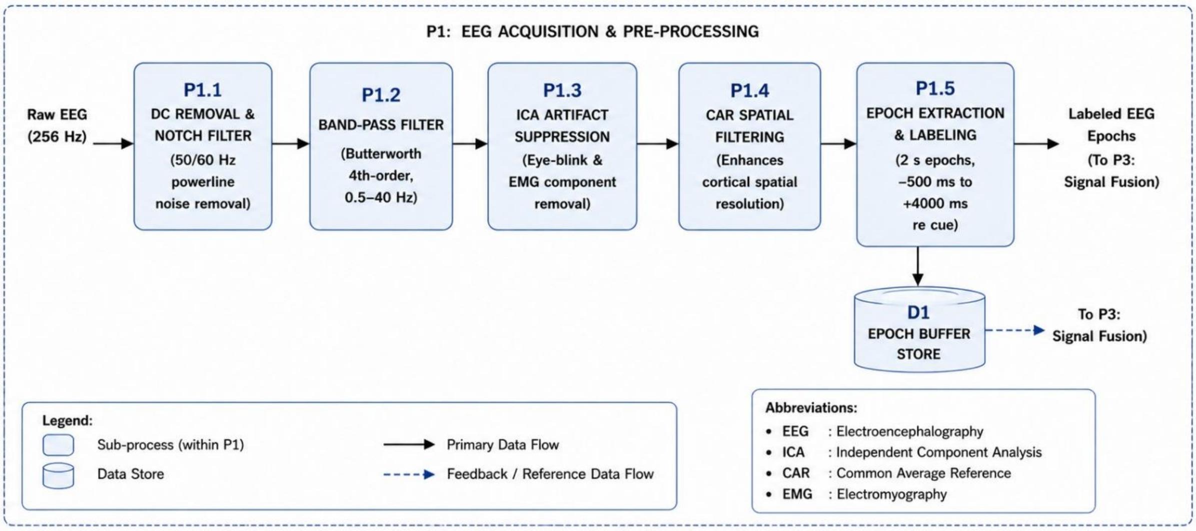

Figure 5.5.3: DFD Level 2- Sub –processes within P1(EEG Acquisition and Pre-Processing)

The EEG processing module consists of the following sequential sub-processes:

- **P1.1: DC Removal and Notch Filtering**
  Removes baseline drift and powerline interference (50/60 Hz).
- **P1.2: Band-pass Filtering**
  Applies a Butterworth filter (0.5–40 Hz) to isolate relevant neural frequency components.
- **P1.3: ICA-Based Artifact Suppression**
  Removes artifacts such as eye blinks and muscle activity.
- **P1.4: Common Average Referencing (CAR)**
  Enhances spatial resolution by re-referencing signals across channels.
- **P1.5: Epoch Extraction and Labeling**
  Segments EEG into time-locked epochs relative to motor cues (e.g., −500 ms to +4000 ms).

The processed EEG epochs are then forwarded to the fusion module (P3) for integration with motion data.

5.6 DEPLOYMENT DESIGN

The proposed Neuro-Adaptive Rehabilitation System is designed to operate under a hybrid deployment architecture, supporting both edge-based and cloud-assisted modes. This design enables real-time processing on the patient device while optionally leveraging cloud infrastructure for storage, analytics, and remote monitoring.

5.6.1 DEPLOYMENT ARCHITECTURE

The deployment architecture follows an edge-centric processing model, where all latency-sensitive computations are executed locally on the patient device, while non-critical tasks are offloaded to the cloud.

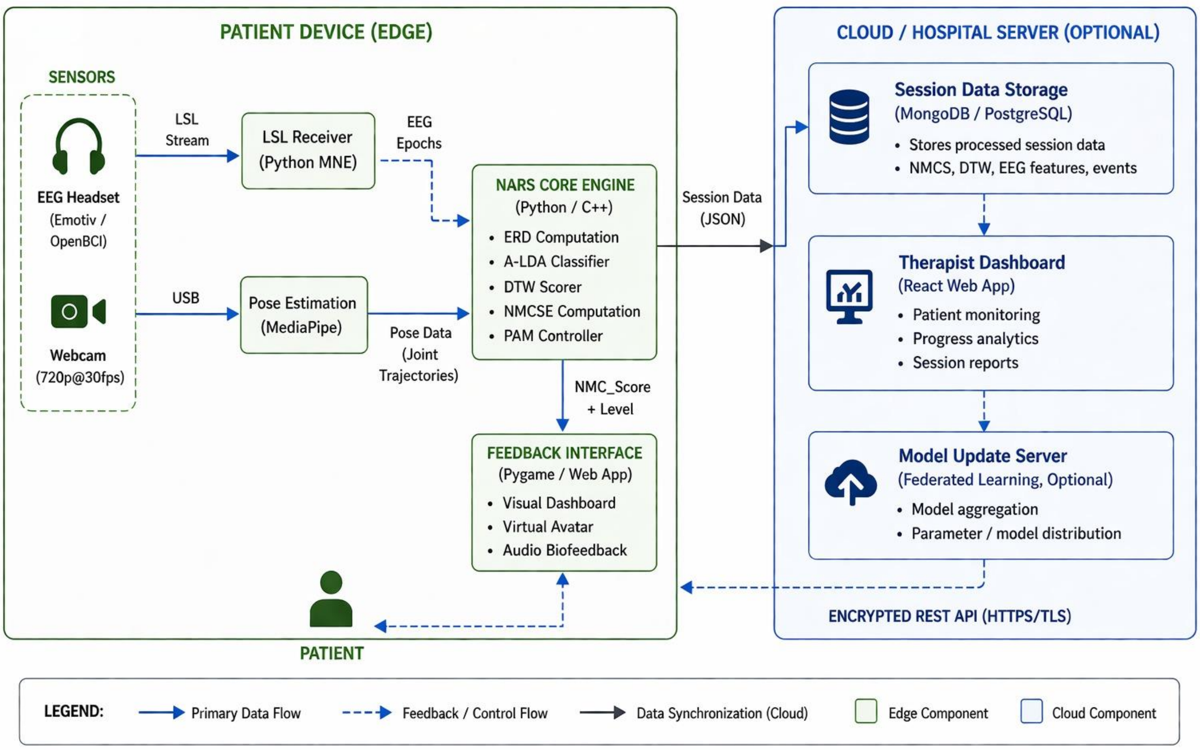

Figure 5.6.1: Deployment Architecture

1. Edge Layer (Patient Device):

    - **EEG Acquisition Module**: EEG signals are captured using a wearable headset and streamed via the Lab Streaming Layer (LSL) protocol to a local receiver.
    - **Motion Capture Module:** A webcam captures body motion, which is processed using pose estimation (e.g., MediaPipe) to obtain joint trajectories.
    - **NARS Core Engine:** The core processing unit performs:

- ERD computation
- LDA-based classification
- DTW-based motion scoring
- NMCS computation
- Performance Adaptation (PAM)

- **Feedback Interface:** The system provides real-time feedback through:
    - visual display (dashboard/avatars)
    - audio cues
    - session summaries

2. Cloud Layer (Backend)

The cloud layer is used for non-real-time tasks, including storage and monitoring:

- **Session Database:** Stores session data such as EEG features, DTW scores, and NMCS values.
- **Therapist Dashboard:** Allows clinicians to monitor patient progress remotely.
- **Model Update Service (Optional):** Supports periodic updates or federated learning without transmitting raw EEG data.

Communication between edge and cloud is performed using secure, encrypted APIs.

5.6.2 DATA FLOW AND INTEGRATION

The deployment pipeline can be summarized as:

Sensors (EEG + Camera) → Edge Processing → Feedback → Cloud Sync

1. Local Processing Flow

    a. EEG and motion data are acquired simultaneously.
    b. Signals are processed locally by the NARS engine.
    c. NMCS and difficulty level are computed.
    d. Immediate feedback is delivered to the user.

2. Cloud Synchronization

After each session:

Session Data→JSON Serialization→Encrypted Transmission→Cloud Storage

Only processed data (not raw EEG streams) is transmitted to reduce bandwidth and preserve privacy.

5.6.3 API INTEGRATION

The system exposes RESTful APIs for communication between edge devices and cloud services.

1. Data Upload API

POST /session/upload

Payload includes:

- NMCS score
- EEG-derived features
- DTW scores
- session metadata

2. Therapist Access API

Returns session history and performance summaries.

GET /patient/id/sessions

5.7 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed Neuro-Adaptive Rehabilitation System offers several advantages in terms of real-time processing capability, multimodal integration, adaptability, and deployment flexibility. These advantages arise from the combined use of EEG-based intention detection and motion-based performance assessment within a unified framework.

1. **Real-Time Edge Processing:**
   The system employs an edge-centric design, where all latency-sensitive computations (EEG processing, motion analysis, and NMCS computation) are performed locally. This ensures low-latency feedback, reduced network dependence, and reliable operation in home environments.

2. **Multimodal Integration**
   By combining EEG-based intention and motion-based execution through NMCS, the system enables comprehensive assessment, including detection of intention–execution mismatch and improved rehabilitation feedback.

3. **Adaptive Feedback Mechanism**
   The Performance Accommodation Mechanism (PAM) dynamically adjusts task difficulty based on recent performance, enabling personalized progression and stable adaptation without abrupt changes.

4. **Robustness to Temporal Variability**
   The use of DTW allows tolerance to variations in movement speed, enabling accurate comparison of trajectories without strict timing constraints.

5. **Modular Architecture**
   The system is structured into independent modules (EEG processing, motion analysis, fusion, and control), allowing easier maintenance, scalability, and component-level improvements.

6. **Hybrid Edge–Cloud Deployment**
   The architecture supports both standalone edge operation and optional cloud integration for storage and remote monitoring, enabling flexible deployment in clinical and home settings.

7. **Privacy Preservation**

   Only processed data is transmitted to the cloud, while raw EEG remains on-device, reducing data exposure and enhancing privacy.

8. **Interpretability**

   Structured outputs such as ERD, DTW score, and NMCS provide interpretable metrics, facilitating clinical understanding and validation.

5.8 HYPOTHESIS AND EVALUATION

This section defines the statistical hypotheses used to evaluate the effectiveness of the proposed Neuro-Adaptive Rehabilitation System, along with the quantitative performance metrics employed to assess classification and regression components of the system.

5.8.1 HYPOTHESIS FORMULATION

To validate the effectiveness of the proposed multimodal rehabilitation framework, hypothesis testing is conducted by comparing the system's performance against a baseline or random performance level.

1. **Null Hypothesis (H_0)**

The null hypothesis assumes that the proposed system does not provide any significant improvement in detecting motor intention or evaluating movement quality. Formally,

$$H_0 : \mu_{model} = \mu_{baseline}$$

where μ_{model} represents the performance metric (e.g., classification accuracy or NMCS correlation) of the proposed system, and $\mu_{baseline}$ represents the corresponding metric under baseline conditions.

This implies that any observed improvement is due to random variation rather than the effectiveness of the proposed model.

2. **Alternative Hypothesis (H_1)**

The alternative hypothesis states that the proposed system provides a statistically significant improvement over the baseline:

$$H_1 : \mu_{model} > \mu_{baseline}$$

This corresponds to a one-tailed test, where the objective is to demonstrate that the proposed system outperforms existing or naive approaches in terms of intention detection and movement evaluation.

3. **Significance Testing**

The hypotheses are evaluated using appropriate statistical tests (e.g., paired t-test or non-parametric alternatives), with a predefined significance level α (typically α=0.05). The null hypothesis is rejected if the p-value satisfies:

$$p < \alpha$$

5.8.2 PERFORMANCE METRICS

The EEG intention model is evaluated as a binary classifier (intent vs. no intent, or focused vs. unfocused).

1. **Precision**

Precision measures the proportion of correctly predicted positive instances among all predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP denotes true positives and FP denotes false positives.

2. **Recall**

Recall measures the proportion of actual positive instances that are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN denotes false negatives.

3. **F1 – Score**

The F1-score is the harmonic mean of precision and recall:

$$\text{F1} = 2 \,.\, \frac{(\text{Precision} \,.\, Recall)}{\text{Precision} + Recall}$$

This metric balances the trade-off between false positives and false negatives.

5.8.3 REGRESSION METRIC

The motion quality model and NMCS output are evaluated using regression-based metrics.

- **Mean Squared Error (MSE):** Mean Squared Error quantifies the average squared difference between predicted and target values:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

where y_i is the ground truth value and $\hat{y}_i$ is the predicted value.

5.8.4 INTERPRETATION OF METRICS

- High **precision** indicates low false positive rate
- High **recall** indicates effective detection of true intent
- High **F1-score** reflects balanced classification performance
- High **F1-score** reflects balanced classification performance

Together, these metrics provide a comprehensive evaluation of both neural intention detection and motion quality assessment.

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